QuickPhone Repair Tracking System: Physical Data Flow Diagrams

MIS 4173 Sprint 3 · Team 4 · Method: Dennis, Wixom, Roth & Haley, *Systems Analysis and Design*, 8th ed., Ch. 4 and Ch. 9

Contents

1. Context diagram: the whole system as process 0 and its five external entities.
2. Level 0 diagram, complete: nine processes, seven data stores, five entities, each drawn once.
3. Level 1 diagrams for 1.0, 2.0, 3.0, 4.0, 5.0, 6.0, 7.0 and 9.0. Process 8.0 Report Repair Status is a read-only primitive with no Level 1.

The slide deck shows the same Level 0 in four slide-sized parts; they contain no flows that are not on the complete diagram.

How the logical DFDs were made physical (Fig. 9-1)

1. **Implementation references.** Each process names how it runs (bottom band, e.g. "Django · bench ThinkPad"). Each data store is a group of tables in the PostgreSQL database, named "PostgreSQL: …" with its tables listed. Each data flow names its medium.
2. **Human–machine boundary.** As in Figs. 9-2 and 9-4, a dashed red line around each person who uses the system's screens: Front Desk, Technician, Manager and Owner. The two manual steps in 6.0 (check photo ID, take payment on the unlinked card machine) are human action and sit inside the Front Desk's line. The Customer has no line: like the Customer Service Department in Fig. 9-2, the customer only receives a printout and an email and never uses the system's screens.
3. **System-related components.** 9.5 Import Inventory File (one-time CSV load) with its import file and error report; audit tables job_status_history and inventory_txn inside D3 and D4.
4. **Data elements.** Each flow carries the elements listed for its logical counterpart in the data flow dictionary (models document, Appendix B), under the physical field names of the ERD, plus system fields such as time stamps and who made the change.

How to read a flow name

Following Figs. 9-2 and 9-4, every line is one data flow with one noun name that says what it is and how it travels:

Name pattern	Meaning
… `Input Form`	A form a person fills in on screen.
Plain data name	What the system shows a person on screen: a list, an alert, a confirmation. The name says what the data is, not the medium; lists end in *List* (for example *Ready Jobs List*).
… `Printout`, … `Email`, … `File`, … `Report`	Printed paper, the email sent from the store's account, an uploaded file, a generated report.
`PostgreSQL Record: …`	Rows read from, or a new row added to, the store the line touches.
`PostgreSQL Record Update: …`	Existing rows changed in the store the line touches.

A Level 0 flow that groups several screens (for example *Check-In Input Form*) splits into named parts on its Level 1; the table that follows lists every split, so each Level 1 balances with Level 0. Lines run only horizontally and vertically and cross at right angles (Ch. 4).

Assumptions and open items

The implementation is a Django (Python) web application with one PostgreSQL database for both stores, run in a browser on the stores' Lenovo ThinkPads, chosen provisionally in the Sprint 3 alternative matrix (415 of 500, against 350 for a repair-shop subscription); the choice reverses if nobody will support the system after December 9. The owner's clarifications of October 23 are drawn; the next page lists where.

Owner clarifications of October 23: where they are drawn

Each answer from the owner, and the processes and flows that changed because of it.

Owner's answer	Where it is drawn
1. Pickup notification is sent by a click from the Technician or authorized staff, never automatically	5.4 Send Email Message takes Send Approval Input Form (Front Desk, Manager, Owner) or Send Notification Input Form (Repair Technician, new on the context, the complete Level 0 and Level 0 part 3). The system fills in the customer's address from D1, so the Technician never sees it. **To confirm:** this reverses the September 21 confirmation #6 (Technicians may not send email); the newer instruction is drawn.
2. A Manager approves every purchase order; no Owner approval	7.3 Record Purchase Order Approval: Pending Purchase Orders List and Purchase Order Approval Input Form now go to and from the Store Manager; the Store Owner keeps only 8.0. **To confirm:** whether a Manager may approve an order they created.
3. Low-stock alerts, technician part requests, Manager-only purchasing, no automatic ordering	7.1 List Parts to Order sends the Low Stock Alert List with the open part requests; 4.4 Record Part Request takes a Part Request Input Form from the Technician. Orders are placed off-system by the Manager (7.4); no process orders automatically and no Technician flow reaches 7.0.
4. Manual assignment; technicians may assign themselves	3.1 Select Repair Job lists the technician's jobs with the unassigned jobs and appointments (D2, new on 3.0) and takes a Job Claim Input Form. The Technician Assignment Input Form at 2.4 is optional; the Front Desk assignment screens are also open to Manager accounts.
5. No diagnostic fee when a device cannot be repaired	6.3 Record Payment passes a Returned Unrepaired job on as a Settled Job with no payment row.
6. Managers set each part's reorder point	Unchanged: the Part Catalog Input Form at 9.3 carries the reorder point per store; 7.1 alerts when stock reaches it.
7. Intake fields; optional passcode, needed only for camera repairs, stored securely	2.4 Create Repair Job takes an optional Passcode Input Form and stores it encrypted with the job. **To confirm:** who may see a passcode.
8. Stores use Lenovo ThinkPad computers	Context and bench process bands name the stores' ThinkPads.
Fixed at the same time	4.0: the Stock Shortage line from 4.2 now ends on 4.4 (it ended on 4.3). The balancing table that follows now uses the PostgreSQL flow names shown on the diagrams. Flow names: a form a person fills in is an *Input Form*, and what the system shows a person is named for its data with no medium word (it was called an Output Form, then a Screen), e.g. Ready Jobs List, Part Request Confirmation; a few names were changed to say what the data is, e.g. Part Request Confirmation.

Balancing: Level 0 flows and their Level 1 parts

Every flow into or out of a Level 0 process appears on its Level 1 diagram, under the same name or split into the named parts below (SAD 8e Ch. 4, balancing).

Level 0 flow	Level 1 flows
1.0 Schedule Appointment	
Appointment Request Input Form	Requested Date Input Form; Booking Input Form; Appointment Change Input Form
Appointment Schedule	Available Times; Booking Confirmation; Change Confirmation
PostgreSQL Record: Store Hours and Technicians	Same name on Level 1
PostgreSQL Record: Existing Customer	Same name on Level 1
PostgreSQL Record: New Customer	Same name on Level 1
PostgreSQL Record Update: Appointment	PostgreSQL Record: New Appointment; PostgreSQL Record Update: Appointment Change
PostgreSQL Record: Booked Appointments	Same name on Level 1; also PostgreSQL Record: Appointment to Change
2.0 Register Repair Job	
Check-In Input Form	Phone Search Input Form; Customer Details Input Form; Device Intake Input Form; Passcode Input Form; Technician Assignment Input Form
Matching Customers	Same name on Level 1
Device Tag Printout	Same name on Level 1
Claim Ticket Printout	Same name on Level 1 (to the Customer)
PostgreSQL Record: Matching Customers	Same name on Level 1
PostgreSQL Record Update: Customer and Device	PostgreSQL Record Update: Customer; PostgreSQL Record: New Device
PostgreSQL Record: Today's Appointment	Same name on Level 1
PostgreSQL Record Update: Appointment Checked In	Same name on Level 1
PostgreSQL Record: Technician Roster	Same name on Level 1
PostgreSQL Record: Store Contact	Same name on Level 1
PostgreSQL Record: New Repair Job	Same name on Level 1
3.0 Record Diagnosis	

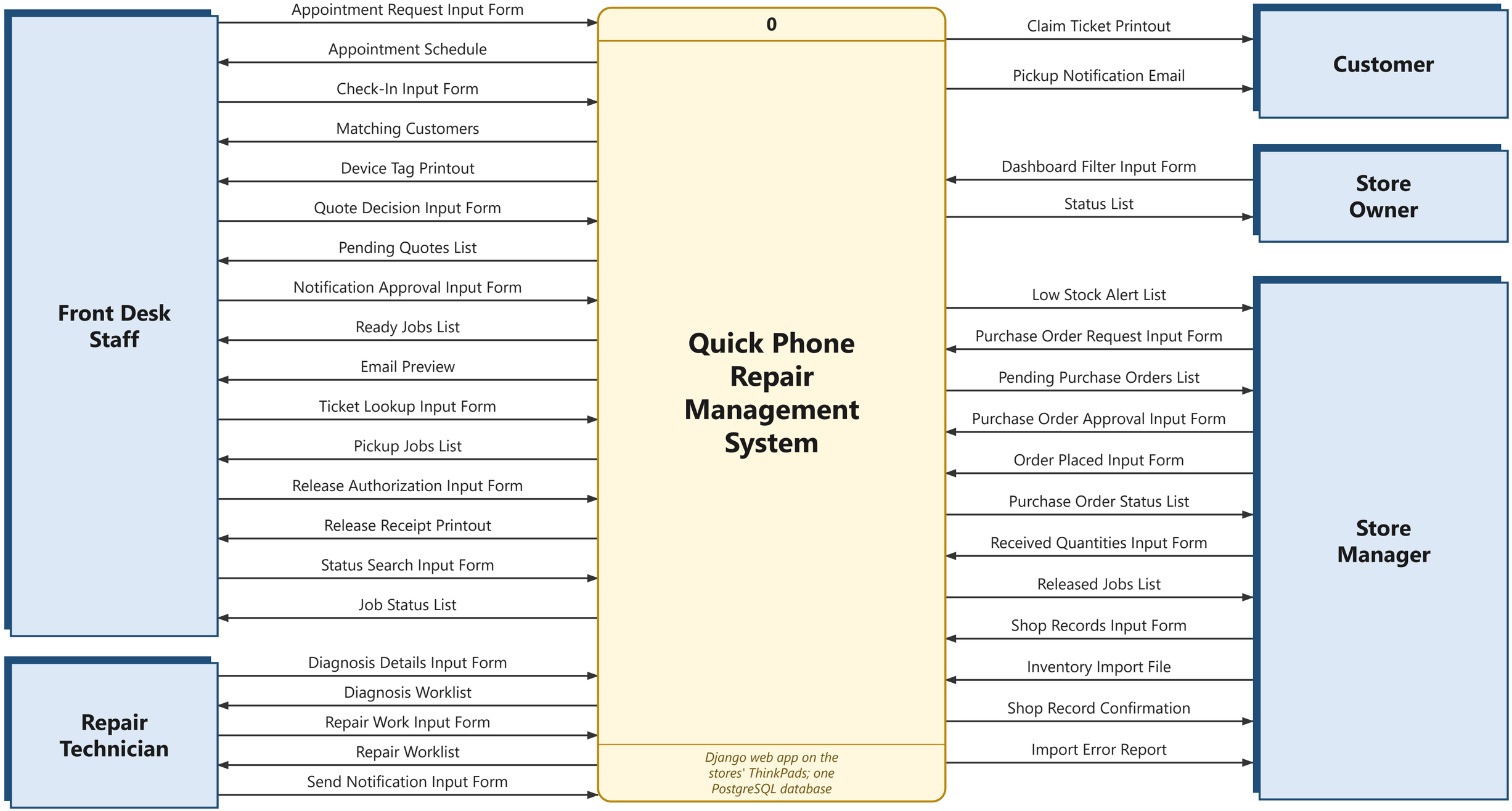
Diagnosis Details Input Form	Diagnosis Job Selection Input Form; Job Claim Input Form; Diagnosis Findings Input Form; Labor Amount Input Form
Diagnosis Worklist	Job Queue; Part Availability
Quote Decision Input Form	Same name on Level 1
Pending Quotes List	Same name on Level 1
PostgreSQL Record: Diagnosis Queue	PostgreSQL Record: Open Jobs; PostgreSQL Record: Pending Quote
PostgreSQL Record Update: Diagnosis and Quote	PostgreSQL Record Update: Job Claimed; PostgreSQL Record Update: Diagnosis Findings; PostgreSQL Record: Part Needed; PostgreSQL Record: New Quote; PostgreSQL Record Update: Quote Decision
PostgreSQL Record: Unassigned Appointments	Same name on Level 1
PostgreSQL Record Update: Appointment Claimed	Same name on Level 1
PostgreSQL Record: Part Availability	Same name on Level 1
4.0 Record Repair Work	
Repair Work Input Form	Repair Job Selection Input Form; Parts Used Input Form; Part Request Input Form; Repair Completion Input Form
Repair Worklist	Repair Job Details; Part Request Confirmation
PostgreSQL Record: Job in Repair	Same name on Level 1
PostgreSQL Record Update: Repair Work	PostgreSQL Record: Parts Used; PostgreSQL Record: Part Request; PostgreSQL Record Update: Repair Completion
PostgreSQL Record: Stock on Hand	Same name on Level 1
PostgreSQL Record Update: Stock Deduction	Same name on Level 1
5.0 Send Pickup Notification	
Notification Approval Input Form	Ready Job Selection Input Form; Final Cost Input Form; Send Approval Input Form
Send Notification Input Form	Same name on Level 1 (from the Repair Technician, to 5.4)
Ready Jobs List	Same name on Level 1
Email Preview	Same name on Level 1
Pickup Notification Email	Same name on Level 1
PostgreSQL Record: Ready Jobs	PostgreSQL Record: Unnotified Ready Jobs; PostgreSQL Record: Approved Quote Total
PostgreSQL Record Update: Notice Sent	PostgreSQL Record Update: Final Cost; PostgreSQL Record Update: Email Sent
PostgreSQL Record: Customer Email Address	Same name on Level 1

PostgreSQL Record: Store Pickup Details	Same name on Level 1
6.0 Record Device Release	
Ticket Lookup Input Form	Same name on Level 1
Release Authorization Input Form	Photo ID and Claim Ticket (to manual 6.5, which gives ID Check Input Form); Payment Input Form (from manual 6.6); Signature Input Form
Pickup Jobs List	Same name on Level 1; also Customer Identity and Amount Due (read by staff doing 6.5 and 6.6)
Release Receipt Printout	Same name on Level 1
PostgreSQL Record: Customer Name	Same name on Level 1
PostgreSQL Record: Job Awaiting Pickup	Same name on Level 1
PostgreSQL Record Update: Payment and Release	PostgreSQL Record: New Payment; PostgreSQL Record Update: Job Released
7.0 Process Purchase Orders	
Purchase Order Request Input Form	Same name on Level 1
Order Placed Input Form	Same name on Level 1
Received Quantities Input Form	Same name on Level 1
Low Stock Alert List	Same name on Level 1
Purchase Order Status List	Same name on Level 1
Released Jobs List	Same name on Level 1
Purchase Order Approval Input Form	Same name on Level 1
Pending Purchase Orders List	Same name on Level 1
PostgreSQL Record: Jobs Awaiting Parts	PostgreSQL Record: Open Part Requests; PostgreSQL Record: Jobs Waiting on PO
PostgreSQL Record Update: Awaiting-Parts Jobs	PostgreSQL Record Update: Job Part Order; PostgreSQL Record Update: Jobs Released
PostgreSQL Record: Low-Stock Parts	Same name on Level 1
PostgreSQL Record Update: Stock Received	Same name on Level 1
PostgreSQL Record: Open Purchase Orders	PostgreSQL Record: POs Pending Approval; PostgreSQL Record: Approved POs; PostgreSQL Record: Ordered PO Lines
PostgreSQL Record Update: Purchase Order	PostgreSQL Record: New Purchase Order; PostgreSQL Record Update: PO Decision; PostgreSQL Record Update: PO Placed; PostgreSQL Record Update: PO Received
PostgreSQL Record: Supplier for Order	Same name on Level 1

9.0 Maintain Shop Records	
Shop Records Input Form	Staff Account Input Form; Store Details Input Form; Part Catalog Input Form; Supplier Input Form
Shop Record Confirmation	Staff Account Confirmation; Store Details Confirmation; Part Catalog Confirmation; Supplier Confirmation
Inventory Import File	Same name on Level 1
Import Error Report	Same name on Level 1
PostgreSQL Record: Staff and Stores	PostgreSQL Record: Staff Accounts; PostgreSQL Record: Store Details
PostgreSQL Record Update: Staff and Stores	PostgreSQL Record Update: Staff Account; PostgreSQL Record Update: Store Details
PostgreSQL Record: Parts Catalog	Same name on Level 1; also PostgreSQL Record: Existing Part Names
PostgreSQL Record Update: Parts Catalog	Same name on Level 1; also PostgreSQL Record: Imported Stock
PostgreSQL Record: Supplier List	Same name on Level 1; also PostgreSQL Record: Supplier Names
PostgreSQL Record Update: Supplier List	Same name on Level 1

QuickPhone — Context Diagram (physical)

The whole system as one process and the five external entities it serves; flows match the Level 0 diagram



QuickPhone — Level 0 DFD (complete) (physical)

All nine processes, seven data stores and five external entities, each drawn once. Each line is one data flow; its name sits beside the process it leaves or enters.

Human–machine boundary (around each person who uses the system's screens)

Front Desk Staff

Quote Decision Input Form
Pending Quotes List

Appointment Request Input Form
Appointment Schedule

1.0 Schedule Appointment
Django booking screen

PostgreSQL Record: Booked Appointments
PostgreSQL Record Update: Appointment
PostgreSQL Record: Existing Customer
PostgreSQL Record: New Customer
PostgreSQL Record: Store Hours and Technicians

Status Search Input Form
Job Status List

8.0 Report Repair Status
Django dashboard

PostgreSQL Record: Customer Contact
PostgreSQL Record: Store List
PostgreSQL Record: Job Status History
PostgreSQL Record: Stock Summary
Dashboard Filter Input Form
Status List

Ticket Lookup Input Form
Pickup Jobs List
Release Authorization Input Form
Release Receipt Printout

6.0 Record Device Release
Django; card machine not linked

PostgreSQL Record: Customer Name
PostgreSQL Record: Job Awaiting Pickup
PostgreSQL Record Update: Payment and Release

Check-In Input Form
Matching Customers
Device Tag Printout

2.0 Register Repair Job
Django + label printer

PostgreSQL Record: Today's Appointment
PostgreSQL Record Update: Appointment Checked In
PostgreSQL Record: Matching Customers
PostgreSQL Record Update: Customer and Device
PostgreSQL Record: Technician Roster
PostgreSQL Record: Store Contact
PostgreSQL Record: New Repair Job

Notification Approval Input Form
Ready Jobs List
Email Preview

5.0 Send Pickup Notification
Django → store email (SMTP)

PostgreSQL Record: Customer Email Address
PostgreSQL Record: Store Pickup Details
PostgreSQL Record: Ready Jobs
PostgreSQL Record Update: Notice Sent

Pickup Notification Email
Send Notification Input Form
Claim Ticket Printout

Customer

D2 PostgreSQL: Appointments
appointment

D1 PostgreSQL: Customers & Devices
customer, device

D7 PostgreSQL: Staff & Stores
store, employee

D3 PostgreSQL: Repair Jobs
repair_job, job_status_history, quote, payment, job_part

D4 PostgreSQL: Parts Inventory
part, store_part_stock, inventory_txn

D6 PostgreSQL: Suppliers
supplier

D5 PostgreSQL: Purchase Orders
purchase_order, po_line

PostgreSQL Record: Unassigned Appointments
PostgreSQL Record Update: Appointment Claimed
PostgreSQL Record: Diagnosis Queue
PostgreSQL Record Update: Diagnosis and Quote
PostgreSQL Record: Part Availability

3.0 Record Diagnosis
Django · bench ThinkPad

Diagnosis Details Input Form
Diagnosis Worklist

PostgreSQL Record: Job in Repair
PostgreSQL Record Update: Repair Work
PostgreSQL Record: Stock on Hand
PostgreSQL Record Update: Stock Deduction

4.0 Record Repair Work
Django · bench ThinkPad

Repair Work Input Form
Repair Worklist

Repair Technician

PostgreSQL Record: Staff and Stores
PostgreSQL Record Update: Staff and Stores
PostgreSQL Record: Parts Catalog
PostgreSQL Record Update: Parts Catalog
PostgreSQL Record: Supplier List
PostgreSQL Record Update: Supplier List

9.0 Maintain Shop Records
Django admin + CSV import

Shop Records Input Form
Shop Record Confirmation
Inventory Import File
Import Error Report

PostgreSQL Record: Jobs Awaiting Parts
PostgreSQL Record Update: Awaiting-Parts Jobs
PostgreSQL Record: Low-Stock Parts
PostgreSQL Record Update: Stock Received
PostgreSQL Record: Supplier for Order
PostgreSQL Record: Open Purchase Orders
PostgreSQL Record Update: Purchase Order

7.0 Process Purchase Orders
Django; orders placed off-system

Purchase Order Request Input Form
Order Placed Input Form
Received Quantities Input Form
Low Stock Alert List
Purchase Order Status List
Released Jobs List
Purchase Order Approval Input Form
Pending Purchase Orders List

Store Manager

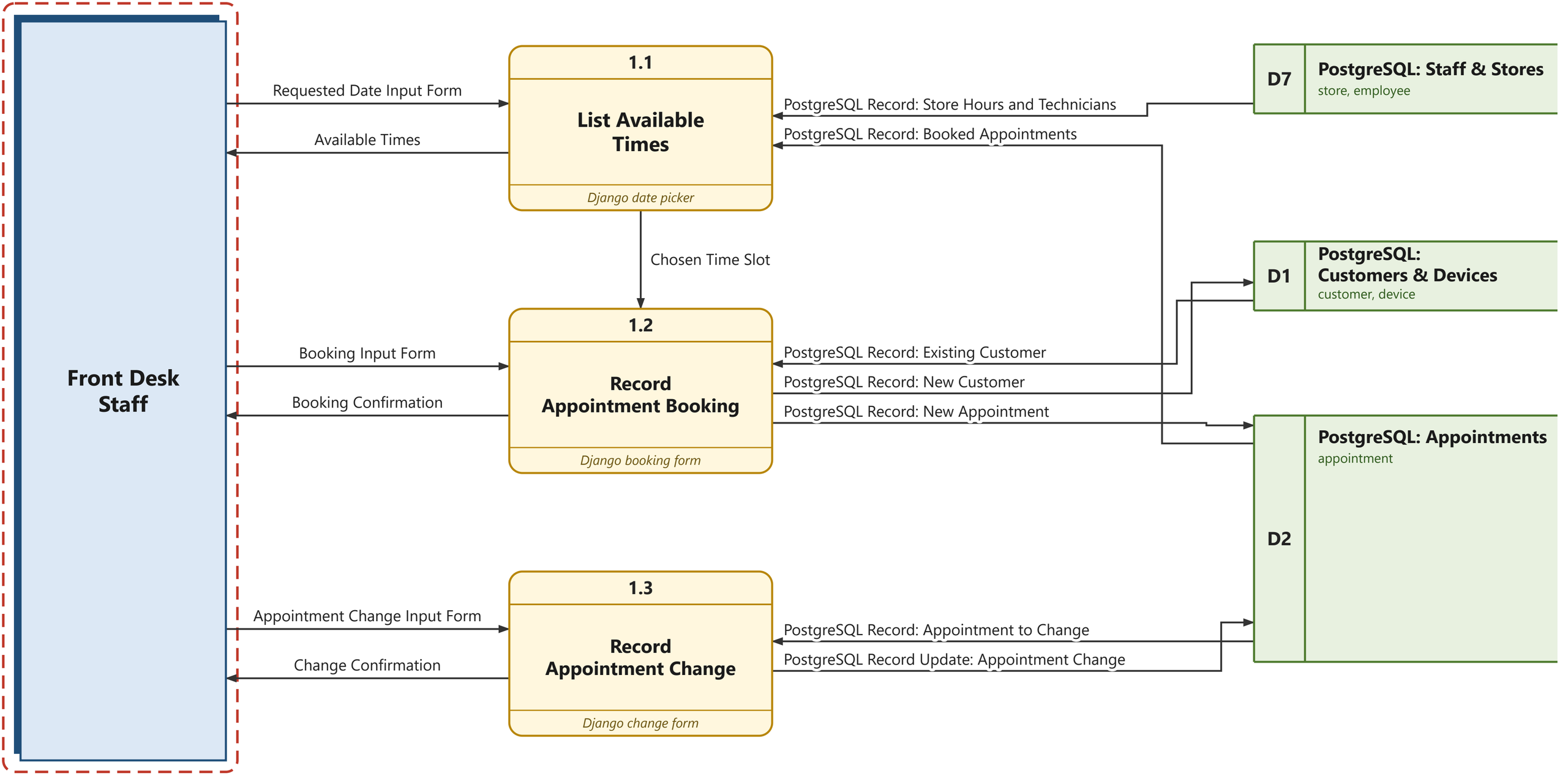
Store Owner

Physical DFD symbols (Gane and Sarson)
External entity
Automated process
Manual process
Data store (ERD entities listed)
Data flow (one way)
Human–machine boundary

Bottom band of a process = how it is implemented and by whom.

QuickPhone — Level 1 DFD: 1.0 Schedule Appointment (physical)

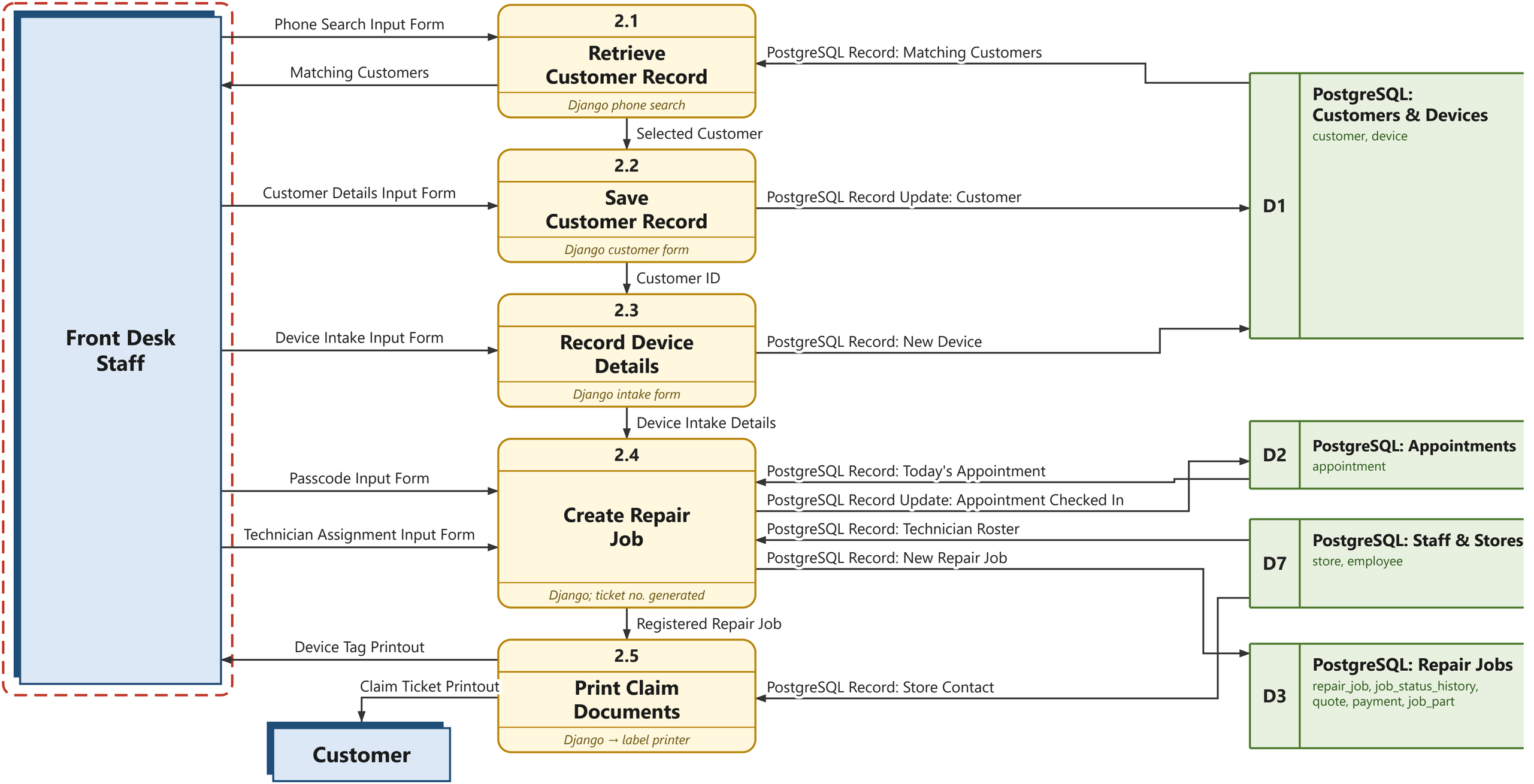
Optional appointment booking (FR01, UC-12). A walk-in needs no appointment; 2.0 pre-fills a job from one when it exists.



QuickPhone — Level 1 DFD: 2.0 Register Repair Job (physical)

Human–machine boundary (around each person who uses the system's screens)

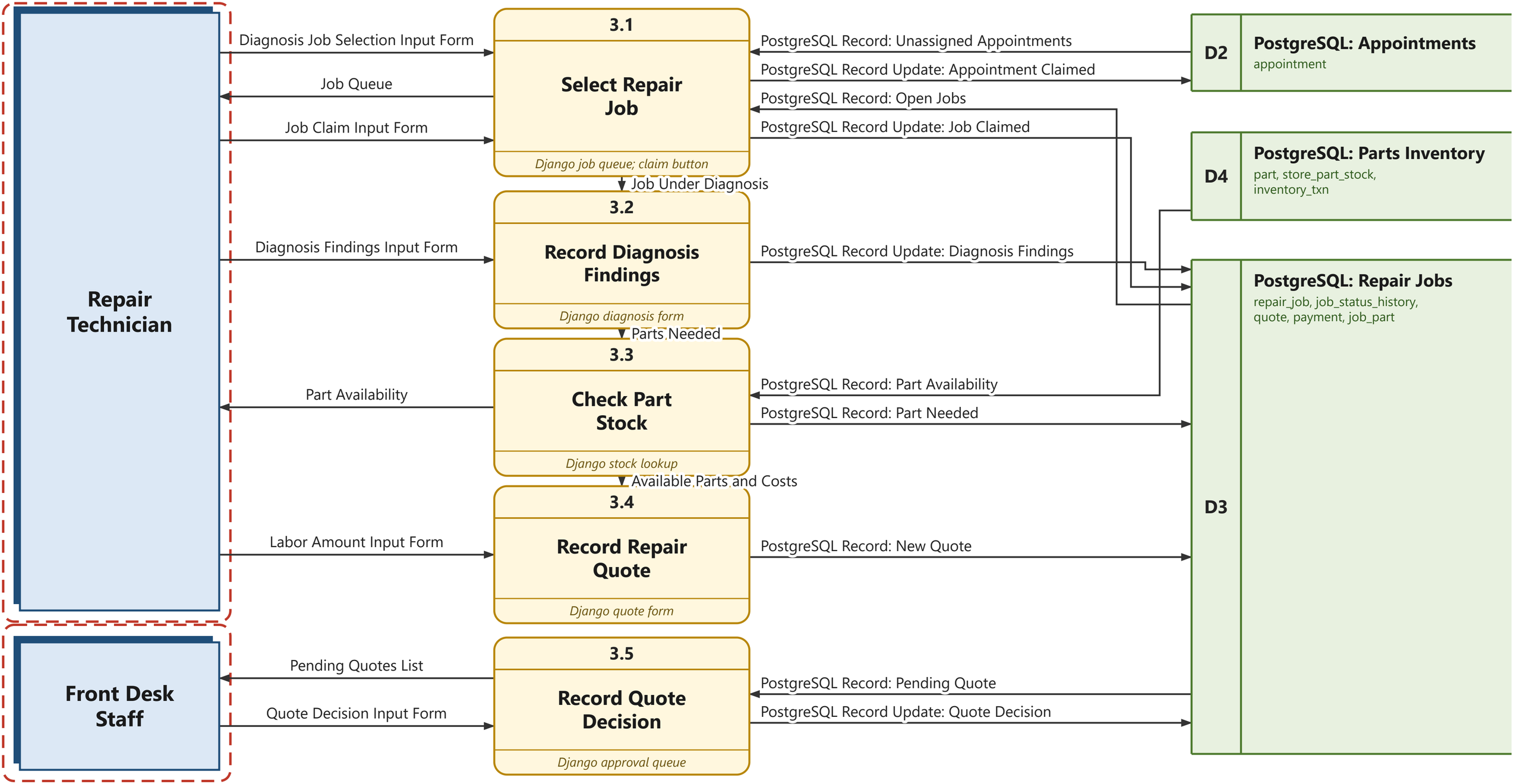
From UC-01 and FR02–FR04. The passcode is optional (needed for camera repairs) and stored encrypted; the technician is optional.



QuickPhone — Level 1 DFD: 3.0 Record Diagnosis (physical)

From UC-02 and UC-03 (FR04–FR07). 3.1 lists the technician's jobs and the unassigned jobs and appointments they may claim.

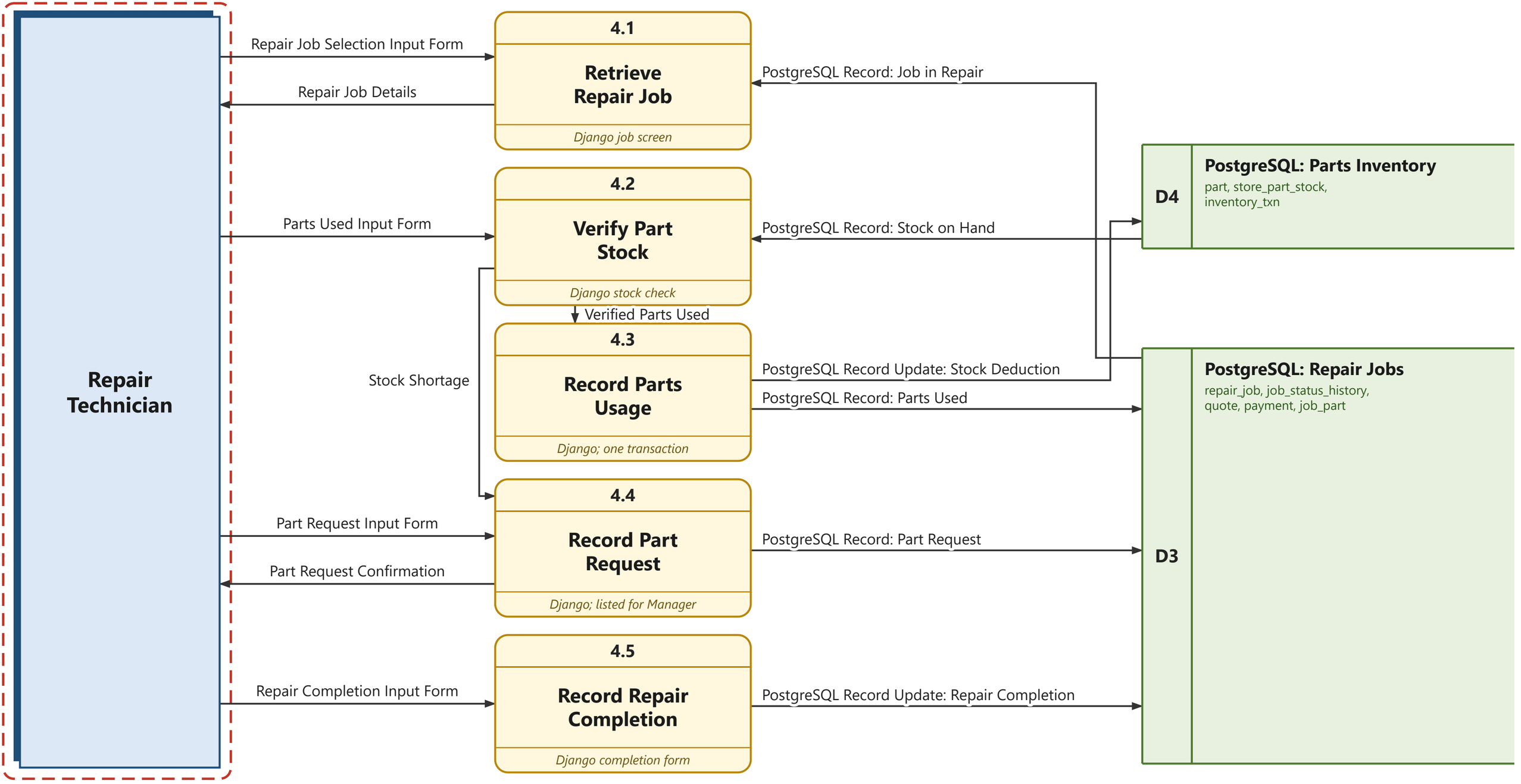
Human–machine boundary (around each person who uses the system's screens)



QuickPhone — Level 1 DFD: 4.0 Record Repair Work (physical)

Human–machine boundary (around each person who uses the system's screens)

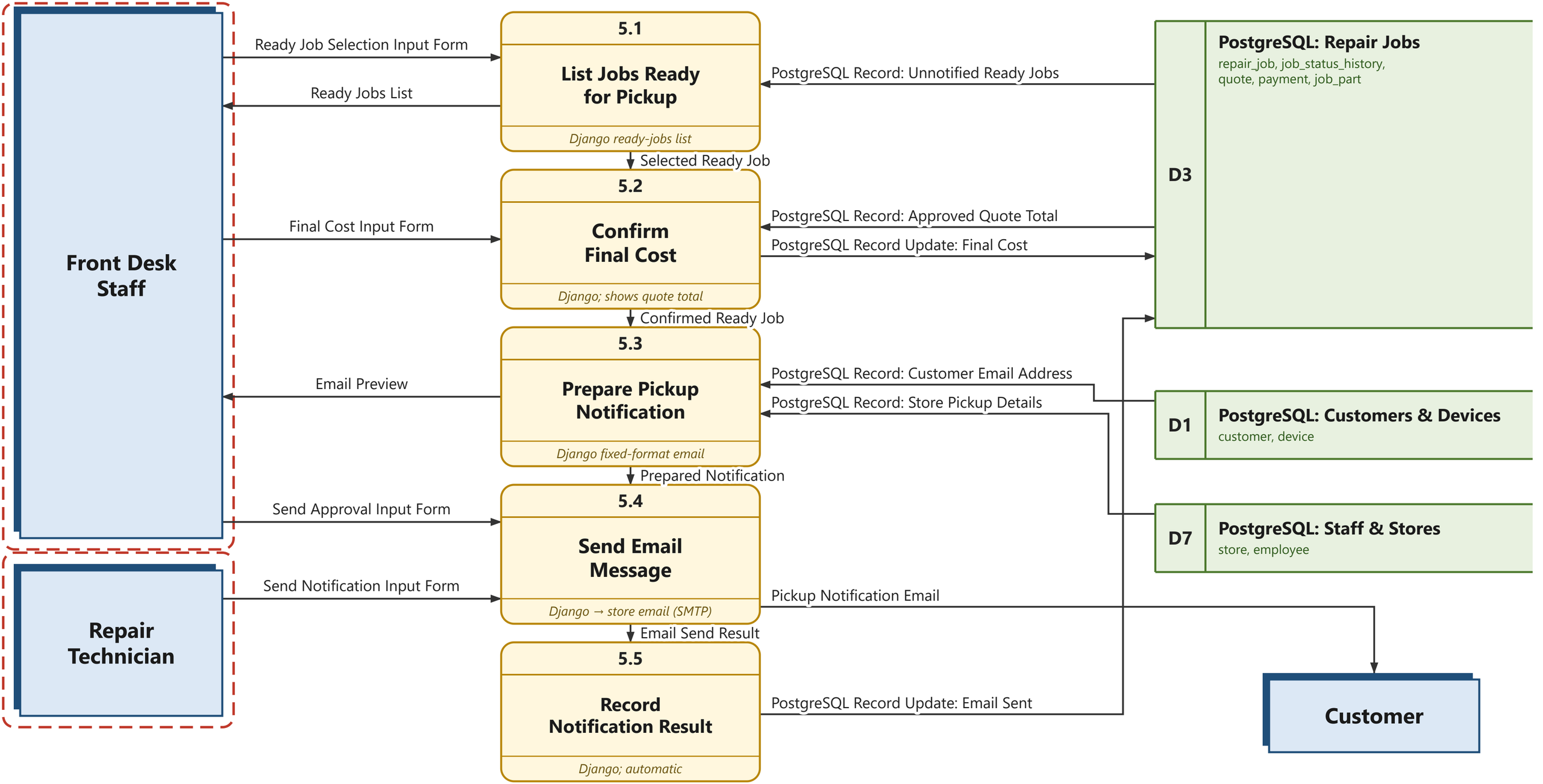
From UC-04 (FR06, FR07). A part used is deducted from stock automatically; a part request goes to the Manager, who orders it.



QuickPhone — Level 1 DFD: 5.0 Send Pickup Notification (physical)

Human–machine boundary (around each person who uses the system's screens)

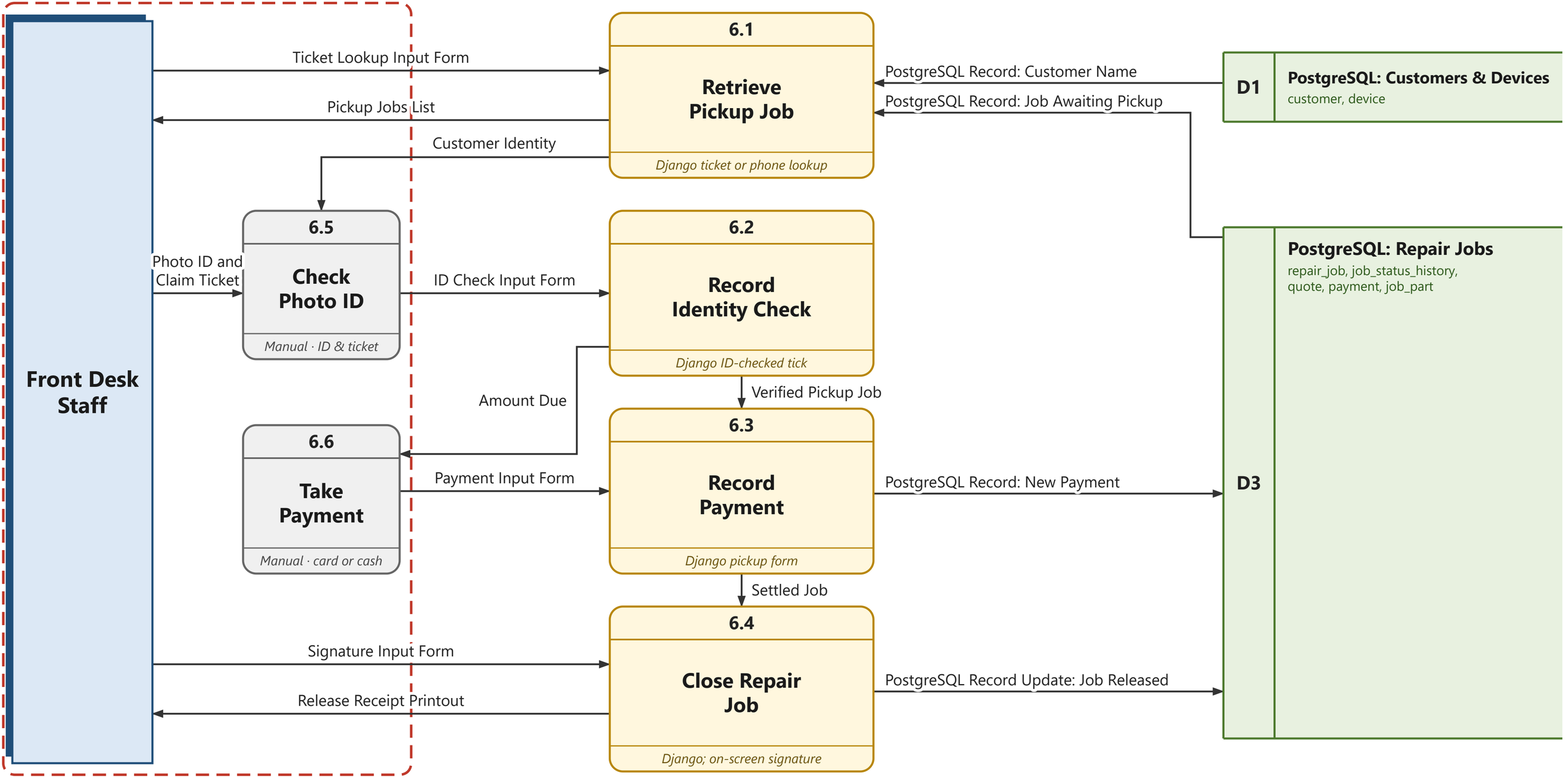
From UC-05 (FR12). Sent only on a click: Send Approval (staff) or Send Notification (Technician); the Technician never sees the address.



QuickPhone — Level 1 DFD: 6.0 Record Device Release (physical)

Human–machine boundary (around each person who uses the system's screens)

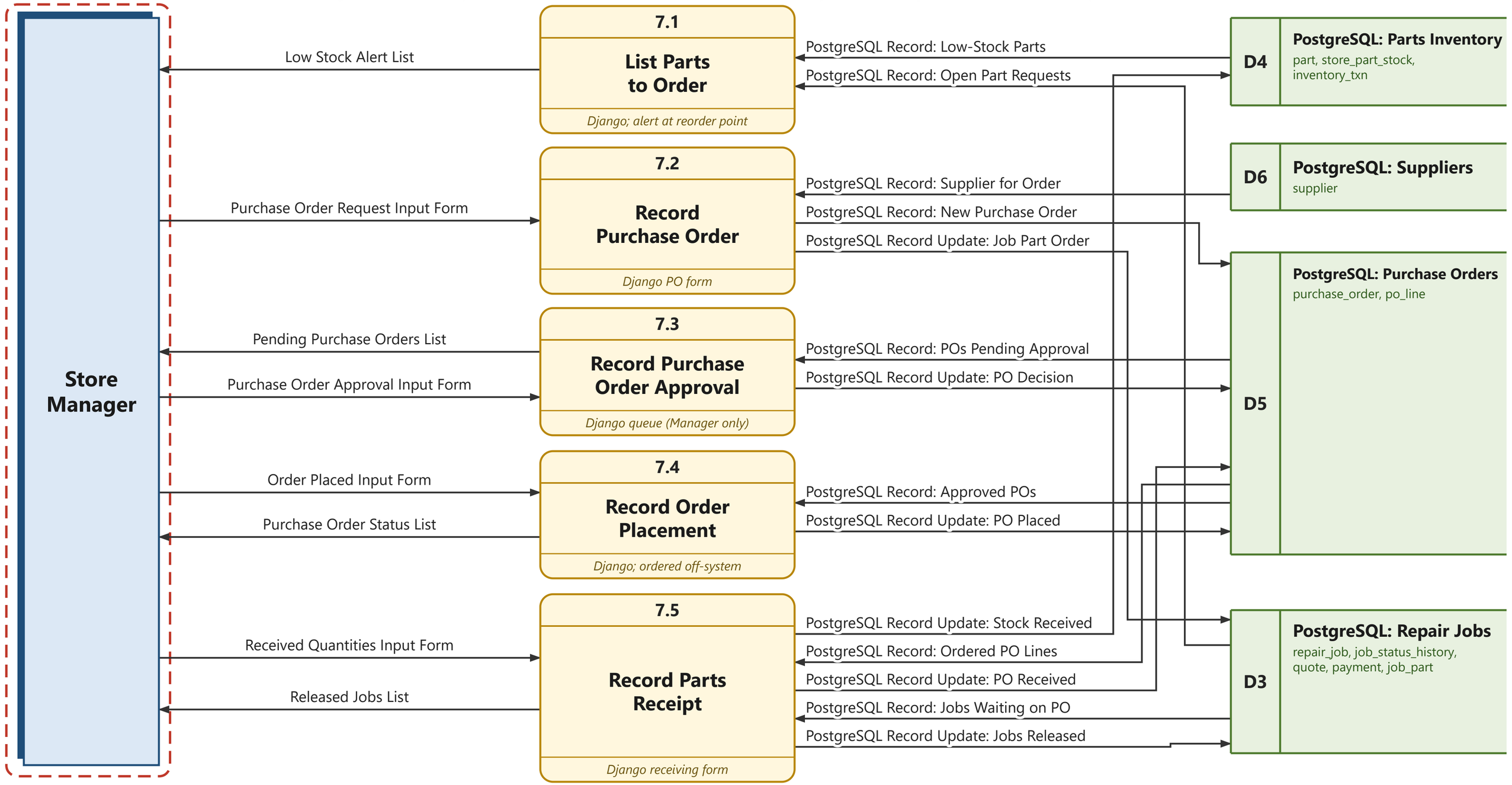
From UC-06 (FR08). 6.5 and 6.6 are manual. An unrepairable device owes nothing (no diagnostic fee), so 6.3 settles it with no payment.



QuickPhone — Level 1 DFD: 7.0 Process Purchase Orders (physical)

Human–machine boundary (around each person who uses the system's screens)

From UC-08 to UC-10 (FR09, FR10). A Manager approves every order and places it with the supplier outside the system; nothing is ordered automatically.



QuickPhone — Level 1 DFD: 9.0 Maintain Shop Records (physical)

From UC-07 and UC-11 (FR09, FR11, FR14). 9.5 adds the one-time CSV import of starting inventory as a batch process.

